INDEX

240701026

NAME : Akhil Kumaran. STD.:_____ SEC.:_____ ROLL NO.: 240701026 SUB.:_____

Computer network obsn.

S. No.	Date	Title	Page No.	Teacher's Sign/Remarks
1)	20/7/26	Basic networking	8/10	
2)	27/8/26	Study of different of Network Cable	7/10	
3)	30/7/26	Packet Trace tool Installation and use. Interface.	8/10	
4)	14/8/26	Setting up a local Area Network (LAN)	8/10	
5)	14/8/26	Experiment on Packet Capture tool Wireshark	7/10	
6)	21/9/24	Hamming Code - Error detection and Correction	8/10	
7)	24/9/26	Sliding window	8/10	
8)	20/9/26	Nmap Live Hosting. Discovery ARP, ICMP. TCP / UDP	9/10	

S. No.	Date	Title	Page No.	Teacher's Sign/Remarks
9)	28/9/20	Implementation of Subnetting in cisco packet tracer	9/10	
10).	29/9/20	a) Interfacing with routers in cisco b) Configuration of internet using multi router DHCP Server	9/10	
11)	30/9/20	a) Stimulate Static Routing Configuration in cisco b) Stimulate RIP using CISCO Packet Tracer	9/10	
12)	1/10/20	End to End Communication At transport layer	10/10	
13.	4/10/20	Implement your own ping program	10/10	
14)	6/10/20	Packet Sniffing using RAW sockets	10/10	
15)	9/10/20	Analysis of web logs using web alizer	10/10	

Completed 11/11/20

Exp.

Exp No: 01

Basic Networking Commands.

Aim.

Study of various networks Commands used in linux and ~~linux~~ windows

1) arp -a.

It shows the Computer IP address and the MAC address of devices on your local network

Output:

Interface 192.168.56.1 ... 0x7.

Internet Address	Physical address	Type
192.168.56.255	ff-ff-ff-f-ff	static
224.0.0.2.	01-00-5e-00-02	static

2) host name:

It is the simplest of all TCP/IP Commands

Output

hp-hemalatha-51

3) ip config/all:

It displays detailed Configuration information about your TCP/IP Connection including routers, gateway, DNS, DHCP, and type of ethernet adapter

output : Window IP Configuration.

Hostname ..... hp-hemaladha-ST

Primary DNS Suffix.

Node Type .... : Mixed.

IP Routing Enabled .... No

WINS Proxy Enabled : NO.

4. Nbtstat -a.

This command help solved problem with Net BIOS name resolution.

Output

Displays Protocol statistics and current TCP/IP connections using NBT.

-a (adapter status) List the remote machine name.

-A (Adapter Status) List the remote machine name table given ctc IP address.

-c (cache) List, NBT's cache of remote names. and their IP address.

-n (names) List, local Net BIOS names.

ayTxt() "> Click me <b v Host>

let Value = thi

document. get Elem

5) netstat: Displays active network connections routing table and network statics.

Proto.	Local Address	Foreign Address	State.
TCP	127.0.0.1497	hp-akhil/49679	Established
TCP	127.0.0.149677	hp-akhil 49678	
TCP	127.0.0.1.49680	hp-akhil 49681.	
TCP	127.0.0.1 49681	hp-akhil -49691.	

6) nslookup: used to find DNS information that is IP address of a domain and DNS records.

Output

Default Server : unknown.

Addres : 172.16.8.1

7). Pathping.

unique to windows and Combines the ping. and Tracert Commands.

Output

Pathping [-g host -cost] [-h- maximum-hops]

[-i addr.] [-n] [-p period] [-v num]

8) Ping [Packet Internet groper].

Commands is the most way to test Connecting between two nodes.

LINUX Commands

ip:

The ip command is used to manage IP address netwo

Output

ip [options] object {command | help}

ip [-force] -batch filename

(a) to show ip address assigned to an interface

(b) ip addr del 192.168.1.254/24 dev enp3

(c) ip config.

It was a stab in many system admin. tool for configuring and trouble shooting between network

Output.

enp3s0 : flags = 4163 <UP, BROADCAST.

RUNNING, MULTICAST > mtu 1500

inet 6 fe80::a66:11:255

inet eth0 00:27:0e:13:f6:d9 (ethernet)

Rx packet 33 5629 bytes

Rx Cum dropped 17.

(3) mtr: MTR is a Program with command line interface that serve as a network diagonise and troubleshooting tool

hagTxt() "> Click me Host>

Output

My traceroute [v0.87]

Host	Packet Loss%	Snt	last	Avg	Best	Wrst	StDev
1-::: 1	0.0%	145	0.2	0.1	0.0	10.6	0.9

(4) Tcpdump

The 'tcpdump : you don't have permission to

Capture on that device

(5) Ping:

Ping is a tool that verifies IP level connectivity to another TCP/IP computer by sending Internet Control Message.

Output

PING google.com (142.200.182.78) 56(84) bytes of data 64 bytes from kenna-arc-dc.

= 1 ttl = 112 time = 93.9 ms

Result

Thus the basic networking and Linux commands are learned and the verification

11/X 8/10

Pg Exp. 2.

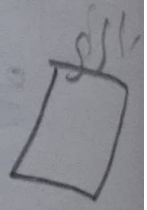
Study of different types of Network Cables

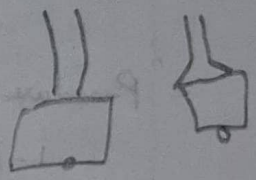
Aim.

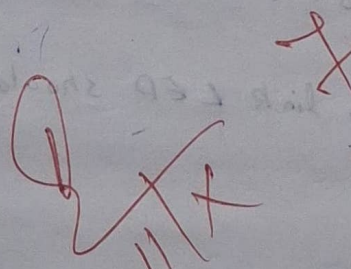
Study of different types of network Cables.

a) understand different types of cable used in networking are.

1) Unshield twisted Pair (UTP) Cable
2) shield Twisted Pair (STP) Cable
3) Coaxial Cable
4) Fibr optic cable

Cable type.	Category.	Maximum data Transmission	Advantages	Apps.	Image.
UTP	Category 3	10 bps	Advantages . Cheaper in cost . Easy to install. as they have a smaller outer dimen. Disadvantage . More Prone to EMI and Noise.	10Base-T Ethernet Fast Ethernet, gigabit Ethernet Fast Ethernet gigabit Ethernet	
	Category 5	upto 100 Mbps.			
	Category 5e.				
STP	Category 6/6a	10. Gbps	Adv. * shielded. * Faster than UTP.	Great for 10.G Ethernet Coax. Jun in data Cent. gigabit Eth	
SSTP	Category 7.	10 Gbps.	Dis. * Expensive * grabs in table.		

Coaxial Cable	RG-6 RG-58 RG-11	10-100 mbps.	Adv. * bandwidth * Low loss. bandwidth Disadv. * Limited distance * Cost.	
fibre optic Cable	Single mode. multimode.	1000 Gbps	Adv. * high speed * high Bandwidth Disadv * Expensive. * Requires skilled installation	



Exp NO3 Packet Tracker Tool Installation and User Interface

Aim

To understand environment of Cisco Packet Tracker to design simple network

Procedure

1. From the network Component box, click & drag-and-drop

a. 4 generic PCs & 1 HUB

b 4 generic PCs & 1 switch

2 Click on Connections.

a Click on Copper wire.

b select on 1 PC and Connect it to HUB using the Cable. The link LED should glow in green., indicating link is up

c. similarly Connect 4 PCs & Switch

Click on the PDU and drag and drop it to one of the PC and then drop it on another PC Connected to HUB.

4. Observe the flow of PDU from source PC to destination PC by selecting the Real time.

5. Repeat Step 3 to Step 4 for the PCs connected to the switch

6. Observe how HUB & switch are forwarding the PDU.

Result

Thus the connection between the server and switch or hub was made successfully.

Exp 4. Setting up a Local Area Network

Dt: 27/8/26

Aim

To design, Configure and verify a 4-PC Local Area Network (LAN) using a central Switch. Stable IP addressing, ping connectivity tests and HTTP resource sharing.

Equipment Required

Equipment	Quantity	Specification
Personal Computer	4	ethernet (NIC)
Network Switch	1	8/16/24 port
ethernet Cable	4	Straight-through

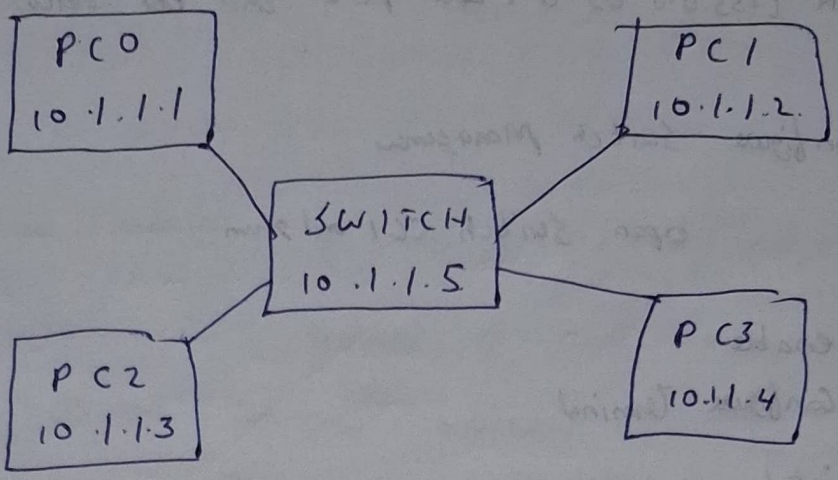
Theory

A Local Area Network (LAN) connects devices within a limited area to share resources. This experiment uses a star Topology where host connects to a central Layer 2 switch. Devices communicate by sharing an IP network range defined by their subnet mask.

Det

Network diagram

Star Topology LAN



IP Configuration table

Device	IP Address	Subnet Mask	Connected to
PC0	10.1.1.1	255.0.0.0	Switch Port Fa 0/1
PC1	10.1.1.2	255.0.0.0	Switch Port 0/4
PC2	10.1.1.3	255.0.0.0	Switch Port Fa 0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa 0/5
Switch	10.1.1.5	255.0.0.0	VLAN

Implementation Steps

1. Place Devices.

Drag 4PC, 1 server and 1 Switch (2960) to the Workspace

2. Cable Connections.

Connect PCs (Fa0/1 to Fa 0/4).

and Server (Fa0/5) to Switch using Copper Straight Cables.

3. Assign Host IPs

Set Static IP addresses and Subnet mask (255.0.0.0) on all PC's and the Server.

4. Configure Switch Management

open Switch CLI and run.

```
enable
Configure terminal
interface Vlan.
ip addres 10.1.1.5 255.0.0.0.
No Shutdown.
exit
```

5. Verify Connectivity.

Run Ping 10.1.1.2 from PC0 Command Prompt to verify network link.

6. Demonstrate Resource Sharing

enable HTTP on Server (10.1.1.10).

open Client PC Web browser and navigate to [http://10.1.1.10]

7. observe Packet flow.

Switch to Simulation mode to trace ARP and ICMP/HTTP frame forwarding across ports.

~~Output:~~

Result.

The LAN lab experiment was successfully built, configured, and verified for IP connectivity and client-server resource sharing.

11/11 8/10

Exp.No: 05 Experiment on Capturing Tools.

Date: Wire Shark

Aim.

To understand the features of wireshark as a packet capture tools and analyze protocol encapsulation at various layers of a Protocol Stack.

Tools Required:

PC Connected to LAN, Wireshark Software.

Theory:

Wireshark is a passive is a passive network packet analyzer that capture packets into three viewable panes:

* Packet List
* Packet Detail
* Packet Byte

Procedure.

1. general. Capture

Open wire Shark, Select the active interface, set Capture limit to 100 packets and click start.

2. TCP/UDP filters :-

Type tcp (or) udp in the display filter bar, inspect headers and generate a Flow graph.

3. ARP Filters

Type arp in the display filter bar to view address resolution requests and replies.

4. DNS Filters:

Type dns in the display filter bar to observe domain name queries and open the flow graph.

5. HTTP Filters:

Type icmp (or) ip in the display filter bar to view ping requests/replies frames.

7. DHCP filters

Type dhcp, bootp in the display filter bar to analyze IP bar packet

Result

Thus the network Packet analyzer tool Wireshark is captured and analysed using wireshark and the communication flow.

Exp No: 06 Hamming Code - Error detection and Correction

Aim.

To write hamming code for error detection and Correction

Code:

```
def hamming - encryption (data):
k = len (data).
l = 0
while (2**l) <= (k+r+1):
        r += 1
        q = 1
        n = k+r+1
        code = [0]*n
        j = 0
        for i in range. (1,n):
            if (is (i-1) ! = 0):
                code [i] = data [j]
                j += 1
            for j in range (1,n).
                if i & pow.
                pair n = int [code [j]).
        code [pow] = str (pair)
        return (code [1:]
```

```
def hamming code - decryption (data, l, k):
for i in range (n):
    pos = 2 ** i
    pair = 0
    for j in range (1, l+k+1):
        if j < pos:
            pair = pair ^ int (code[j-1])
    if (pair != 0):
        if (data [pos-1] == '0')
            data [pos-1] = '1'
        else:
            data [pos-1] = '0'
    return : join (data).

data = input ("Enter binary data :")
print ("Data after encoding from sender side :")
ndata = input ("enter data after receive):
```

O/P

Enter binary data = 1011

Data after encoding for sender data recieve by

1010111

Data after decoding: 0110011

Result

Thus, the hamming code was detected and corrected successfully.

11/X 8/10

Exp. 2 Sliding Window Protocol.

Aim.

To implement flow control at data link layer using SLIDING WINDOW Protocol simulate the flow frames one node to another.

Sender Program

1. Input window size from the user
2. Input a test message from the user
3. Consider characters per frame
4. Create a frame with following field.
   (frame no, DATA)
5. send the frame
6. wait for acknowledgement from the Receiver
7. Read a file called Receiver.txt
8. Check ACK field for the Acknowledgement no.

Create a receiver file with followings

1. Reads a file called Sender file - Buffer
2. Check the frame no.
3. If the frame no are accepted.

Write the appropriate ACK in the Reciever Buffer file. Else write NACK no in the Reciever Buffer file

Code

```
import time.
import os.
sender_file = "Sender_Buffer.txt"
reciv_file = "Reciever_Buffer.txt"
message = input("Enter text message:")
window_size = int(input("Enter window size:"))
frame = []
frame = []
```

```
print ("In waiting for acknowledgement ...")
time.sleep(2)
if not os.path.exists(receiver_file):
    print ("Receiver buffer not found.")
    break
with open (receiver_file, "r") as f:
    ack = f.read().strip()
    print ("Receiver Response: " ack)
    if ack.startswith ("ACK"):
        ack_no = int (ack.split(":")[1])
        if ack_no == curr:
            print ("Acknowledgement received")
            start = end.
        else:
for no, data in Frame:
    if no != expected:
        with open ("Receiver_Buffer", "w") as f:
            f.write (f"NACK, {expected}")
    else:
        with open ("Receiver Buffer", "w") as f:
            f.write (f"ACK, {no}")
        expected += 1
        time.sleep(1)
    time.sleep(1).
```

Input

Window size : 3

Enter Message : BRICKS

Output - Sender Terminal.

Window size : 3

Enter message : BRICKS.

Sending : [(1, 'B'), (2, 'R'), (3, 'I')]

Received : Ack 3

Sending : [(4, 'K'), (5, 'S')]

Received : ACKS

Transmission Completed.

Output - Receiver Terminal.

Received : [[1, 'B'], [2, 'R'], [3, I])

Received : [[4, 'K'], [5, 'S']]

Result

Thus the code is implemented For Sliding.

Window Protocol.

Exp - 8

N-MAP to Discover live hosts using N-MAP Scan. (ARP, ICMP, TCP/UDP) on the TryHackme platform.

Aim.

To discover live hosts on the network using N-map host discovery. Scans ARP ICMP, TCP/UDP Ping scan. and to understand of the model, and when each one applies.

Introduction.

* Nmap ("Network Mapper") is a Free, open-source network scanning tool created by Gordon Lyon.

* The lab Focus first 4 Questions asked during. pre Commissions.

1. Which services are up? answer in this room.
2. What Services are running on

Nmap can determine whether a host is alive using four different protocols.

* ARP (Link Layer).
* ICMP (Network Layer).
* TCP (Transport Layer).
* UDP (Transport Layer).

ICMP Host Discovery

* Request Several request / Repay Pair that an reveal a live host

* Windows System & many Firewalls blocks ICMP

7. 1 ICMP Echo (-PE)

$ sudo nmap -PE -sn 10.200.6.0/24

Nmap scan report for 10.200.0.1

Host is up (0.095) latency)

Nmap report for 10.200.6.50

Host is up

Nmap done: 256 IP address (3 host) scanned in 3.38 seconds.

ICMP Timestamp (-PP)

$ Nmap -PP -sn 10.200.6.01.024

Nmap scan. report For 10.200.6.50

Host is Up.

Nmap done: 256 IP address (1 host up) scanned in 3.46 second

$ sudo nmap -PA -sn 10.200.6.0

Nmap scan report for 10.200.6.50

Host is up (0.068s latency).

Nmap done: 256 IP addrs (2 hosts scanned) in 33.8
seconds

UDP Ping (-PU).

Sends a UDP packet to a UDP Port.

$ sudo nmap -PU -sn 10.200.6.0/24

Nmap scan report 10.200.6.50

Host is up (0.070s latency)

Nmap scan report for 10.200.6.250

Host is up (0.061s latency)

Nmap done: 256 IP addrs (2 hosts up) scanned
in 35.97 seconds

Result.

Nmap Discovering live host using nmap has been done successfully

Ex - 09.

Implementation of subnetting in CISCO Packet Tracer Simulator

Aim.

* To implement the subnetting in CISCO packet tracer simulator.

Creating A Network Topology:

* The first step in implementing. Class IP subnetting is to create a net.

* Here we will be adding routers, switches PCs.

* To add device, select the device from the bottom left corner & drag it onto the network tool.

Subnetting.

* To Subnet the network address of 192.168.1.0/24, to provide enough space enough space for each address for end device

* This will give us distinct Python addr.
each

Result

Implementing of Subnetting has been done Successfully using cisco packet tracer

Exp-11

Practical - 11 (a).

Static Routing.

Aim.

To Simulate and Configure static routing with Primary and backup routes using Cisco Packet Tracer and verify. Consulting the routes

Procedure

1. Open cisco packet Tracer and Create the given topology with Routers Router 1, Router 2, Switch a.

2) Configure the IP address on all the router interfaces and PC's according to the given - topology

3) Open CLI of Router. and enter Configuration mode.

4) Configure separate host route Routers as a backup route.

5) On Routers, for network 10.0.0.0/8 and 30.0.0.0/8

6) Use traceroute can be removed. using no ip route

11. Static routes can be removed using: no ip route.

12. Verify the routing table again using show ip route static

Result

Thus the routing with Primary, backup routes successfully using GNS3 done.

12.

12. End to end Communication at Transport Layer

Aim.

To implement an echo - client server using TCP/UDP socket using Socket Programming

a) echo - client - server using TCP/UDP

a socket is an endpoint used for communication btw two device. In an echo communication, a client,

1) Create Server Program and Client Program.
2) Import the requires socket library
3) In server create a socket and bind it to an IP address and port.
4) make the server listen to client request
5) Start client and Connect it Server

6) enter new client

7) send msg to server

8) The server receives the msg. and sends echo it

5) Client Server using Socket Prog

D Creat Server using and Client Program. using Socket Programming

2) start Server and bind to an IP. address and port

3) make Server wait for client Communicat

4) Start Client and Connect it to it

5) Enter msg in client

6) Send msg to server

7) Client sending connection to Performance

8) Close the Communication

Result

The end to end Communication was successfully implemented using TCP/UDP Sockets.

Exp 10. Internet working with routers in Cisco Packet Tracker. Simulation

Aim.

To design Configure a simple internetworking a router in cisco packet tracker and task.

Procedure

Step 1:- Configuring Router

* Select the Router and open CLI
* Press enter to start
* Type enable - to activate the Privilaged mode.

Step 2

* Assign IP address to every PC in the network

* Select the PC's go to the desktop and Select IP Configuration and assign an IP address (default gateway, subnet mask

* Repeat the Configuration for all requirements

~~Result~~

Result

Thus, implementation of subnetting in Cisco Packet is done successfully.

13. Implement your own Ping Program

Aim

To implement a simple Ping Program to check the Connectivity between two network devices.

Theory

Ping as a network utility used to check whether a destination device is reachable through network

It sends a request to destination and wait for a reply if received. Connection is established.

Requirement

* Computer
* Python Suitable Programming Environment
* network connection

Procedure

1. Create a new programming for implementing ping operation

2) Import the required socket network

3) Get IP address or host name of the destination.

4) Create a socket for Communication.

5) Send req packet to the destination.

6) Parse the process for received ICMP Packet

Algorithm

1) Start

2) enter destination IP address.

3) Create a Socket

4) Send a ping request

5) wait for response.

6) otherwise declare

7) Stop.

Result:

Thus, the Pig Program was successfully implemented

14. Raw Socket – Packet Sniffing

Aim.

To write a Program using RAW Socket & implement Packet Sniffing

Theory.

A packet Sniffer is a Program used to Capture and examine network Packet traffic through a network

Requirement

* Computer / laptop,
* Python,
* Raw Socket Support

Algorithm

1) Start the Program
2) Create RAW Socket.
3) bind the Socket to ethernet
4) wait for incoming network packets

Procedure.

1) Open Python IDE or terminal

2) Create a python program for Packet Sniffing

3) Import Socket module

4) Create RAW Socket.

5) Read Captured packed data

6) Convert into readable

Result

Thus, RAW socket based packet Sniffing Program was implemented sucessfully

Exp 15 Analysis of web logs using web alizer

Aim.

To analyze the diff types of web logs using web alizer tool

Theory.

Webalizer is a web log analyser tool used to process web server log files and generate

~~Requirement~~ Procedure

* Run the web alizer window will open.
* Select or provide the required log
* Download web log from web if required
* load the web log in web alizer.
* Press Run web alize
* Webalizer process the log file

* Observe the general analysis / report
* Study the different information available in the report

Input

A web server log file

Output

Net log analysis report generated in it website

Result

Thus the web log file was successfully analyzed analyzed using web diz tool.